

PRACTICAL 1 — Inode, Hard Link and Symbolic Link

Create a folder

```
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical$ mkdir Inode_Practical
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical$ cd Inode_Practical
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ pwd
/home/saireddy/COW_Practical/Redirection_Practical/Inode_Practical
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Create a file

```
/home/saireddy/COW_Practical/Redirection_Practical/Inode_Practical
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ echo "Hello Linux Inode" > original.txt
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ cat original.txt
Hello Linux Inode
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Check inode number

```
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ ls -li original.txt
21057 original.txt
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Use stat

```
21057 original.txt
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ stat original.txt
File: original.txt
Size: 18          Blocks: 8          IO Block: 4096   regular file
Device: 8,48     Inode: 21057       Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/saireddy)   Gid: ( 1000/saireddy)
Access: 2026-09-19 05:25:38.548909411 +0000
Modify: 2026-09-19 05:25:27.300909417 +0000
Change: 2026-09-19 05:25:27.300909417 +0000
Birth: 2026-09-19 05:25:27.300909417 +0000
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Create a HARD LINK

```
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ ln original.txt hardlink.txt
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ ls -li
-rw-r--r-- 2 suresh 21057 18 2026-09-19 05:25 original.txt
-rw-r--r-- 2 suresh 21057 18 2026-09-19 05:25 hardlink.txt
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Check hard-link count

```
21057 original.txt 21057 hardlink.txt
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ stat original.txt
File: original.txt
Size: 18          Blocks: 8          IO Block: 4096   regular file
Device: 8,48     Inode: 21057       Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/saireddy)   Gid: ( 1000/saireddy)
Access: 2026-09-19 05:25:38.548909411 +0000
Modify: 2026-09-19 05:25:27.300909417 +0000
Change: 2026-09-19 05:28:19.288898684 +0000
Birth: 2026-09-19 05:25:27.300909417 +0000
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Create a SYMBOLIC LINK

```

suresh@DESKTOP-QDI8C9M: ~/COW_Practical/Redirection_Practical/Inode_Practical$ ln -s original.txt symlink.txt
suresh@DESKTOP-QDI8C9M: ~/COW_Practical/Redirection_Practical/Inode_Practical$ ls -li
total 8
21057 -rw-r--r-- 2 sairedddy sairedddy 18 Sep 19 05:25 hardlink.txt
21057 -rw-r--r-- 2 sairedddy sairedddy 18 Sep 19 05:25 original.txt
21058 lrwxrwxrwx 1 sairedddy sairedddy 12 Sep 19 05:29 symlink.txt -> original.txt
suresh@DESKTOP-QDI8C9M: ~/COW_Practical/Redirection_Practical/Inode_Practical$

```

Use find

```

suresh@DESKTOP-QD18C9M : ~/COW_Practical/Redirection_Practical/Inode_Practical$ find . -inum 123456
suresh@DESKTOP-QD18C9M : ~/COW_Practical/Redirection_Practical/Inode_Practical$ ./original.txt
./hardlink.txt
-bash: ./original.txt: Permission denied
-bash: ./hardlink.txt: Permission denied
suresh@DESKTOP-QD18C9M : ~/COW_Practical/Redirection_Practical/Inode_Practical$

```

Test the links

```
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ echo "New line" >> original.txt
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$ cat hardlink.txt
Hello Linux Inode
New line
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/Inode_Practical$
```

```
suresh@DESKTOP-QDI8C9M: ~/COW_Practical/Redirection_Practical/Inode_Practical$ cat symlink.txt
Hello Linux Inode
New line
suresh@DESKTOP-QDI8C9M: ~/COW_Practical/Redirection_Practical/Inode_Practical$
```

PRACTICAL 2 — mmap() File Reading/Writing

Create input file

```

suresh@DESKTOP-QD18C9M: ~/COW_Practical/Redirection_Practical/Inode_Practical$ cd ..
suresh@DESKTOP-QD18C9M: ~/COW_Practical/Redirection_Practical$ mkdir MMap_Practical
suresh@DESKTOP-QD18C9M: ~/COW_Practical/Redirection_Practical$ cd MMap_Practical
suresh@DESKTOP-QD18C9M: ~/COW_Practical/Redirection_Practical/MMap_Practical$ echo "Hello from mmap system call." > data.txt
suresh@DESKTOP-QD18C9M: ~/COW_Practical/Redirection_Practical/MMap_Practical$ cat data.txt
Hello from mmap system call.
suresh@DESKTOP-QD18C9M: ~/COW_Practical/Redirection_Practical/MMap_Practical$

```

Create C program

```

1  #! /usr/bin/perl
2  #
3  # Copyright (c) 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 2679, 2680, 2681, 2682, 2683
```

```

suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/MMap_Practical$
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/MMap_Practical$ gcc mmap_file.c -o mmap_file
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/MMap_Practical$ ./mmap_file
Original file content:
Hello from mmap system call.

File content after modification:
Memory Mapped I/O!
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/MMap_Practical$ |

```

Check the actual file

```

suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/MMap_Practical$
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/MMap_Practical$ gcc mmap_file.c -o mmap_file
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/MMap_Practical$ ./mmap_file
Original file content:
Hello from mmap system call.

File content after modification:
Memory Mapped I/O!
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/MMap_Practical$ cat data.txt
Memory Mapped I/O!tem call.
suresh@DESKTOP-QDI8C9M :~/COW_Practical/Redirection_Practical/MMap_Practical$ |

```